MIL-S-37499A
21 December 1976
SUPERSEDING
MIL-S-37499
15 March 1976

MILITARY SPECIFICATION

SPLINT, PNEUMATIC, ADULT, SLIP-ON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two double-wall, plastic, pneumatic splints each having a self-sealing valve operated by the insertion of a tubular mouthpiece supplied with each splint. One splint is for the full adult arm and the other for the full adult leg. Both splints are open at both ends.

1.2 Classification. Splints shall be furnished in the following types as specified (see 6.2).

Type I	- Arm
Type II	- Leg

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-676	- Boxes, Set-Up

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D792 Specific Gravity and Density of Plastics by Displacement

ASTM D1238 Measuring Flow Rates of Thermoplastics by Extrusion Plastometer

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 1910 .)

3. REQUIREMENTS

3.1 Material. Materials shall be of the highest commercial quality and shall be entirely suitable for the purpose intended. Material shall contain no ingredients known to be skin sensitizers or irritants.

3.1.1 Plastic for splint. The plastic for the splint shall be clear, transparent, pliant, ethylene vinyl acetate copolymer film having a density of 0.937 ± .09 grams per cubic centimeter (g./cu.cm.) when tested in accordance with 4.3.2 and a melt index of 1.5 ± 0.1 g. per 10 minutes when tested in accordance with 4.3.3. The thickness of the plastic shall be 5.5 to 7.0 mil, when tested as specified in 4.3.1.

3.1.2 Plastic for inflation tube. The inflation tube shall be fabricated from polypropylene.

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3.2 Construction. The arm splint and the leg splint shall be constructed in accordance with Figure 1. The inflation tube shall be supplied attached to splint wall with self-adhesive transparent tape for easy removal. The completed splint shall show no leakage when tested as specified in 4.3.4.

3.3 Instructions for use. Instructions for use shall be furnished, permanently and legibly marked, on the immediate container or contained on a sheet and inserted inside the container (Sec 5.3.1). The instructions shall include appropriate illustrations and the cautionary note.

3.3.1 Government review of instructions. As soon as practicable after award of the contract or purchase order, the contractor shall furnished to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed Instructions for use, and Caution, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary text shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final text without delaying delivery under the contract or purchase order. Should the supplier have furnished acceptable Instructions for use and Caution for the identical item within two years preceding the date of contract and proposes to furnish Instructions/caution identical with those previously accepted, then the requirement for review of the draft may be waived upon request to the contracting officer.

3.4 Workmanship. Workmanship shall be first class throughout. Splints shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three (3) years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the inflation tube and plastic sheeting, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I, II and III (see 4.2.1 and 4.2.2), 4.2.3 and 4.2.4.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I. Unit of product for examination purposes shall be one splint.

TABLE I. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0

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4.2.2 Examination. The splint shall be examined for defects including, but not limited to, those listed in table II.

TABLE II. Classification of defects

Major	
101	Any component missing.
102	Component not item specified.
103	Plastic for bag and for splint not transparent.
104	Visible holes or tears in splint or bag.
105	Plastic bag (unit package) completely not heat-sealed.
106	Two heat-sealed ribs not made on splint.
Minor	
201	Plastic bag for unit package not completely free of excess air.
202	Inflation tube not cut at approximate 45° angle at one end.

4.2.3 Packaging inspection. The inspection of the unit, intermediate package, packing, and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.2.4 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table III, except that a minimum of three samples shall be taken regardless of lot size. Unit of product for test purposes shall be one splint.

TABLE III. Sampling criteria for tests

Component	Characteristic	Requirement	Test Procedure	Inspection Level	AQL
Entire Splint	Dimensions	3.2	----	S-1	1.0
	Thickness of splint plastic film	3.2	4.3.1	S-2	1.0
	Density of splint plastic film	3.1.1	4.3.2	S-2	1.0
	Melt Index of splint plastic film	3.1.1	4.3.3	S-2	1.0
	Splint leakage test	3.2	4.3.4	S-1	1.0

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4.2.5 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.3.1 Thickness of splint plastic film. Samples for test shall be conditioned at a temperature of 73.5° ± 2°F. (23.1° ± 1.1°C.) and a relative humidity of 50 ± 4 percent for a period of not less than 16 hours. The gauge measurement of thickness shall be a dead weight type equipped with a dial graduated to read directly to 0.0001 inch. The presser foot shall be circular with a diameter of 0.25 ± 0.01 inch. The pressure foot and moving parts connected therewith shall be weighted so as to apply a total load of 3 ± 0.1 ounce to the specimen. The pressure foot and anvil surface shall be plane to within 0.0001 inch and parallel to each other within 0.0001 inch.

4.3.2 Density of splint plastic film. Shall be determined according to ASTM Test Method D 792.

4.3.3 Melt Index of splint plastic film. Shall be determined according to ASTM Test Method D 1238.

4.3.4 Splint leakage test. The splint shall be inflated to a pressure of 60 mm of mercury and the valve closed. The splint, thus inflated, shall be observed for air bubble escape after immersion in water, or by the application of a soap solution, immediately after inflation.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Unit of issue. Each splint, of the type specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each splint shall be thoroughly clean, neatly folded to an approximate dimension of 7x8 inches and packaged in a clear air-tight polyethylene bag having a minimum wall thickness of 0.0015 inch. Excess air shall be evacuated from the bag and the closure effected by heat sealing.

5.1.2.2 Intermediate package. Four unit packages containing splints of one type only shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure of the box shall be as specified in the appendix of the applicable box specification.

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5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages (48 units) containing splints of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, water-proofing and strapping shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twelve intermediate packages (48 units) containing splints of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked as specified in MIL-STD-129. In addition, marking shall include name or registered trademark of the manufacturer, instructions for use and the following, in 3/16-inch minimum capital letters:

"CAUTION- USE IN AIRCRAFT REQUIRES MONITORING THE SPLINT PRESSURE TO PREVENT NEUROVASCULAR INJURY. LIMIT INFLATION PRESSURE TO ADEQUATE FRACTURE SUPPORT ONLY." followed by, less prominently, in 1/8-inch high letters:

"The aneroid pressure gauge of a standard sphygmomanometer should be available for monitoring the splint pressure during ascent and descent. (Splints air transported should be inflated to provide adequate support of the fractured limb (below 50 mm Hg). Overinflation is to be specifically avoided.)"

As an alternate, the required marking shall be legibly printed on a separate piece of paper which shall be inserted in the unit package in such a manner as to permit ready identification.

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5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The inflatable plastic splints covered by this specification are for use in emergency treatment of arm or leg fractures.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification.
b. Type (see 1.2).
c. National Stock Number (NSN).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

6.3 The pneumatic splints covered by this specification are listed in the Federal Supply Catalog under the following stock numbers:

National Stock Number	Item Identification
6515-01-010-8762	SPLINT, ARM, Pneumatic, Adult, Slip-on
6515-01-010-8763	SPLINT, LEG, Pneumatic, Adult, Slip-on

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing Activity
Defense Supply Agency - DM

Project No. 6515-2025

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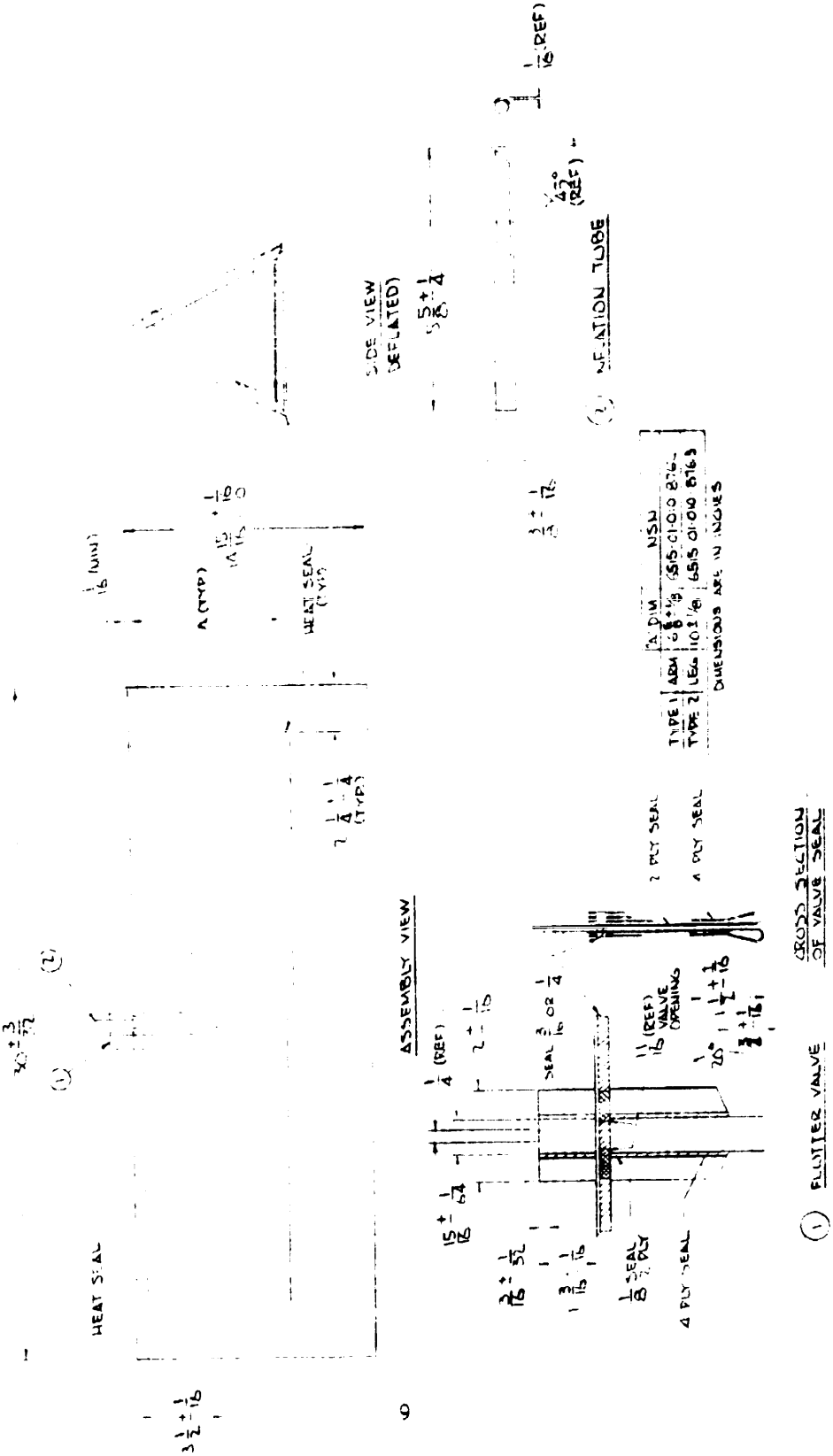


FIGURE 1. Splint, pneumatic, adult, slip-on.

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-S-37499A - SPLINT, PNEUMATIC, ADULT, SLIP-ON

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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